

Hybrid Bio-Inspired and LLM-Based Feature Selection for Medical Diagnosis Problems

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Abstract—Feature selection is a key preprocessing step in machine learning for medical diagnosis, especially when dealing with high-dimensional datasets where irrelevant and redundant features can negatively affect model performance. In clinical applications, reducing the feature space is also important for improving interpretability and computational efficiency.

In this work, we propose a hybrid feature selection framework that combines Large Language Models (LLMs) with bio-inspired optimization techniques. The method integrates semantic feature relevance extracted from an LLM with a bio-inspired intelligence-based multi-objective search strategy. The LLM is used to provide prior knowledge about feature importance, which helps guide population initialization toward semantically relevant regions of the search space.

Experiments are conducted on Parkinson's disease voice data and high-dimensional cancer gene expression datasets. The proposed approach is compared with a baseline PSO-based feature selection method without LLM guidance. Results show that LLM integration improves the quality of selected feature subsets, achieving better classification performance while reducing the number of selected features.

Keywords—feature selection, bio-inspired intelligence, large language models, medical diagnosis, multi-objective optimization, pareto front, non-dominated features.

I. INTRODUCTION

Feature selection (FS) is a critical preprocessing step in machine learning, particularly in medical diagnosis (MD), where datasets are often high-dimensional, heterogeneous, and complex [1] [2]. The presence of redundant, irrelevant, and noisy features can significantly degrade classification performance, increase computational cost, and reduce model interpretability. In clinical applications, interpretability is especially important, as diagnostic decisions must be both accurate and explainable. Therefore, identifying a compact and informative subset of features remains a key challenge in biomedical data analysis.

Traditional FS methods can be broadly categorized into filter, wrapper, and embedded approaches [1] [3]. Although these methods are widely used, they often struggle when applied to high-dimensional medical datasets. Filter methods may ignore feature dependencies, while wrapper and embedded approaches can become computationally expensive and are prone to local optima in large search spaces.

To overcome these limitations, bio-inspired intelligence (BII) algorithms have been widely adopted for FS due to their strong global optimization capabilities and ability to balance exploration and exploitation. Algorithms such as Particle Swarm Optimization (PSO), Genetic Algorithms (GA), Whale Optimization Algorithm (WOA), and Harris Hawks Optimization (HHO) have shown promising results in selecting near-optimal feature subsets for MD tasks [4] [5] [6]. However, despite their effectiveness, these approaches rely mainly on numerical fitness functions and lack semantic understanding of feature relevance.

In particular, most bio-inspired FS methods evaluate candidate solutions solely based on predictive performance metrics, without considering domain knowledge or contextual relationships among features. This limitation becomes more critical in medical datasets such as gene expression data [7] or physiological signals [8], where feature interactions often carry important diagnostic meaning.

Recently, transformer-based Large Language Models (LLMs) have demonstrated strong capabilities in understanding semantic relationships and capturing domain knowledge from textual information [9]. This opens new opportunities for integrating LLMs into optimization pipelines [10]. By leveraging their ability to infer feature importance based on domain-aware reasoning, LLMs can complement traditional metaheuristic algorithms and support Pareto-based selection [11] toward more meaningful feature subsets.

Motivated by this, we propose a hybrid FS framework that integrates LLM-guided reasoning with BII. Specifically, we incorporate LLM-derived feature relevance scores into a multi-objective bio-inspired optimization process to improve both exploration and convergence behavior. The LLM is used to generate semantic feature relevance scores that guide the initialization phase of the optimization process, while the optimization process ensures strong predictive performance.

The proposed method is evaluated on two medical datasets: Parkinson's disease voice measurement data, high-dimensional cancer gene expression data. The experimental results demonstrate that incorporating LLM guidance improves feature subset quality, enhances classification performance, and reduces the number of selected features compared to a baseline bio-inspired approach without LLM assistance.

The main contributions of this paper are summarized as follows:

- A novel hybrid FS framework combining LLM-based semantic reasoning with bio-inspired optimization techniques.
- An LLM-guided population initialization strategy that improves the exploration capability of metaheuristic algorithms in high-dimensional medical datasets.
- A multi-objective Pareto-based FS formulation that integrates semantic priors with traditional performance driven optimization.
- An extensive experimental evaluation conducted on two biomedical datasets (Parkinson's disease and cancer gene expression data) using multiple BII and different LLM configurations.

The remainder of this paper is organized as follows: Section II reviews related work on feature selection, BII methods, LLMs, and hybrid feature selection approaches. Section III presents the background of feature selection problems, bio-inspired optimization, multi-objective optimization (MOO) concepts, and LLM-based feature relevance estimation. Section IV describes the proposed approach, including the overall framework design, LLM-based feature scoring mechanism, and the multi-objective fitness formulation. Section V presents the experimental setup, including datasets, preprocessing, and evaluation metrics. Section VI discusses the results and comparative analysis. Finally, Section VII concludes the paper and outlines future research directions.

II. RELATED WORK

A. FS Methods

FS aims to identify the most informative subset of features from high-dimensional datasets in order to improve predictive performance, reduce computational complexity, and enhance model interpretability. In MD applications, FS plays a critical role because biomedical and multimodal datasets frequently contain noisy, redundant, and irrelevant attributes that negatively affect learning performance [1], [12].

Several surveys have highlighted the importance of FS in biomedical analysis. Remeseiro and Bolón-Canedo [1] reviewed a wide range of FS techniques in medical applications and emphasized the challenges associated with high-dimensional clinical data. Similarly, Tangherloni et al. [3] analyzed FS methods for disease risk prediction and discussed their impact on machine learning-based healthcare systems.

Traditional FS methods are generally categorized into filter, wrapper, and embedded approaches. Filter methods evaluate feature relevance independently of the learning model using statistical or information-theoretic criteria such as Mutual Information, Chi-square, Pearson Correlation, and Information Gain [13], [14]. These approaches are computationally efficient and suitable for large datasets; however, they often ignore dependencies and nonlinear interactions among features.

Wrapper methods rely on predictive models to evaluate candidate feature subsets [15]. In these approaches, the classifier acts as a black-box evaluator, while the search process iteratively identifies feature combinations that

maximize predictive performance. Recursive Feature Elimination (RFE) represents one of the most commonly used wrapper techniques. Although wrapper methods generally achieve higher accuracy than filter methods, they become computationally expensive when applied to high-dimensional biomedical datasets.

Embedded approaches integrate FS directly into the model training process. Methods such as LASSO regularization [16], neural-network-based embedded strategies [17], and tree-based feature importance mechanisms simultaneously perform learning and feature selection. These approaches usually provide a compromise between computational efficiency and predictive capability.

Despite their effectiveness, conventional FS approaches often struggle with nonlinear optimization problems and very large search spaces, motivating the use of metaheuristic and bio-inspired optimization techniques.

B. FS Using BII

BII algorithms have attracted increasing attention for FS due to their strong global optimization capabilities and flexibility in handling complex search spaces [4], [6]. These methods formulate FS as a combinatorial optimization problem in which candidate feature subsets are iteratively updated according to a fitness evaluation function.

Pham and Raahemi [4] provided a comprehensive review of bio-inspired FS algorithms and highlighted their effectiveness in addressing high-dimensional optimization tasks. Similarly, Garg et al. [5] discussed the growing impact of nature-inspired optimization algorithms on single-objective optimization problems between 2019 and 2023.

Evolutionary approaches such as GA employ crossover, mutation, and selection operators to improve population diversity and avoid premature convergence [18]. Altarabichi et al. [18] proposed a fast GA-based FS framework that demonstrated promising performance in classification tasks while reducing feature dimensionality.

Swarm intelligence methods constitute another major category of BII algorithms. Algorithms such as PSO, ACO, WOA, and HHO simulate collective natural behaviors to efficiently explore the search space [6]. Among these approaches, Peng et al. [19] introduced a hierarchical Harris Hawks optimizer for FS and reported strong performance on high-dimensional datasets due to its adaptive exploration-exploitation mechanism.

Bio-inspired methods have shown promising results in several MD applications, including cancer classification [7], EEG signal analysis [8], gene expression analysis [14], and multimodal healthcare systems [12]. Nevertheless, most existing BII-based FS approaches rely exclusively on numerical fitness evaluation metrics and do not incorporate semantic understanding or contextual interpretation of feature relevance.

C. FS with LLM

Recent advances in transformer-based LLMs have opened new research directions for intelligent feature analysis and selection. Unlike traditional statistical approaches, LLMs are capable of capturing semantic relationships, contextual dependencies, and domain-specific knowledge from textual descriptions and structured representations.

Several recent studies explored the integration of LLMs into FS pipelines. Jeong et al. [9] proposed LLM-Select, an LLM-guided FS framework that employs prompting strategies to estimate feature importance and guide feature ranking. Their work demonstrated that semantic reasoning can improve the identification of informative features beyond purely statistical criteria.

Similarly, Li and Xiu [10] introduced LLM4FS, a hybrid strategy combining LLM reasoning with conventional data-driven FS techniques. Their experiments showed that integrating semantic contextualization with traditional statistical evaluation improves feature subset quality and predictive robustness.

In medical applications, LLMs can assist in interpreting biomarkers, physiological signals, and multimodal clinical variables by leveraging domain-aware reasoning. However, current LLM-based FS methods remain limited by computational cost and insufficient optimization mechanisms for exploring large combinatorial search spaces.

D. Hybrid FS Approaches

To overcome the limitations of standalone approaches, recent studies have investigated hybrid frameworks combining metaheuristic optimization with intelligent guidance mechanisms.

Hybrid approaches aim to exploit the exploration capability of bio-inspired optimizers while integrating external knowledge sources to improve search efficiency and solution quality. Existing studies have incorporated machine learning models, neural networks, and semantic reasoning techniques into optimization pipelines [20].

More recently, the integration of LLMs with optimization algorithms has emerged as a promising direction for intelligent FS. Yuan et al. [20] explored an LLM-based FS framework combined with counterfactual explanations for schizophrenia-related functional connectivity analysis. Their work demonstrated the potential of semantic reasoning in biomedical feature analysis.

However, most current hybrid approaches employ LLMs primarily as auxiliary ranking tools and do not fully integrate semantic reasoning throughout the optimization process. In particular, Limited attention has been given to integrating LLM-guided initialization strategies into multi-objective bio-inspired FS frameworks.

Furthermore, research on LLM-guided bio-inspired FS for multimodal and high-dimensional MD datasets, including gene expression and MIMIC-based clinical data, remains relatively limited. Therefore, additional research is needed to develop robust frameworks capable of jointly leveraging semantic reasoning and global optimization capabilities.

III. BACKGROUND

FS aims to identify a subset of informative features from a high-dimensional dataset while preserving the discriminative characteristics required for a learning task. In MD applications, this problem becomes particularly challenging due to the large number of heterogeneous and redundant variables commonly found in biomedical datasets

such as gene expression profiles, physiological signals, and multimodal clinical records [1], [3].

From an optimization perspective, FS is considered an NP-hard combinatorial problem because the number of possible feature subsets increases exponentially with the number of available features [4]. Let: $F = \{f_1, f_2, \dots, f_n\}$ be the original feature space and $F' \subset F$ the selected subset. The objective of FS can be formulated as: maximize $T(F')$ where $T(\cdot)$ represents a target evaluation function such as classification accuracy, weighted F1-score, or a composite optimization criterion.

According to the analysis of Yu and Liu [21], features may be categorized as strongly relevant, weakly relevant, or irrelevant. Strongly relevant features contain essential predictive information and their removal usually degrades model performance. Weakly relevant features may contribute partially or redundantly, whereas irrelevant features mainly introduce noise and increase model complexity. Consequently, an effective FS strategy should preserve discriminative information while minimizing redundancy and dimensionality [1].

In supervised learning, the general FS process usually involves four main stages: subset generation, subset evaluation, stopping criteria, and validation [4]. The search mechanism explores candidate subsets, while an evaluation function measures their predictive quality using a classifier or statistical criterion.

A. Bio-Inspired Optimization for FS

Because exhaustive search becomes computationally infeasible for high-dimensional datasets, BII algorithms have been widely adopted as heuristic optimization approaches for FS [4], [6].

In these approaches, FS is modeled as a population-based optimization problem in which candidate feature subsets evolve iteratively according to a fitness function. The optimization process attempts to maintain a balance between exploration, which searches new regions of the solution space, and exploitation, which refines promising candidate solutions.

A binary solution can be represented as: $x_i \in \{0, 1\}^n$ where each dimension indicates whether a feature is selected or discarded. A commonly used objective function combines classification performance and subset compactness: where $Error(x)$ denotes the classification error, $|x|$ is the number of selected features, and $\alpha \in [0, 1]$ controls the trade-off between predictive performance and dimensionality reduction.

Several BII techniques have been applied to FS problems. Evolutionary approaches such as GA simulate biological evolution using selection, crossover, and mutation operators [18]. Swarm-based methods, including PSO, WOA, ACO, and HHO, imitate collective behaviors observed in nature to efficiently explore large search spaces [6].

Among these methods, PSO remains one of the most widely used optimization techniques due to its simple implementation and low computational complexity. In PSO, each candidate solution is represented by a particle characterized by a position vector and a velocity vector. The

search process updates particles according to personal and global best solutions discovered during optimization.

Despite their effectiveness, BII methods still suffer from several limitations, including premature convergence, sensitivity to parameter tuning, and insufficient semantic understanding of feature relationships [4], [5]. These limitations become more evident in biomedical datasets where interactions between variables may carry important clinical meaning.

B. MOO

FS is naturally formulated as a MOO problem because multiple conflicting objectives must be optimized simultaneously. In most MD tasks, the main objectives are:

- maximizing predictive performance,
- minimizing the number of selected features.

Instead of producing a single optimal solution, MOO generates a set of trade-off solutions known as Pareto-optimal solutions [11].

Given two candidate solutions x_a and x_b , solution x_a dominates x_b if:

- $f_i(x_a) \leq f_i(x_b)$ for all objectives,
- and $f_j(x_a) < f_j(x_b)$ for at least one objective.

The collection of non-dominated solutions forms the Pareto front, which represents different compromises between accuracy and subset compactness. Hypervolume (HV) is commonly used to evaluate the quality of Pareto fronts since it measures the dominated objective space covered by the obtained solutions: Pareto-based optimization is particularly suitable for medical applications because different clinical scenarios may require different trade-offs between interpretability and predictive accuracy.

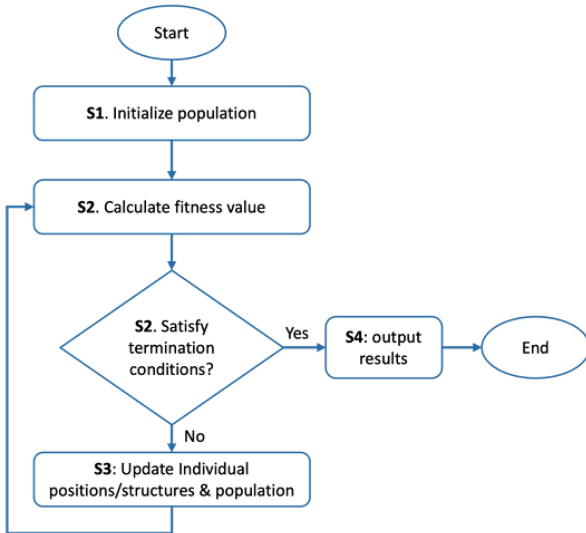


Fig. 1. Framework of BII algorithm. [4]

C. LLM for FS

Recent progress in transformer-based LLMs has introduced new possibilities for incorporating semantic reasoning into FS processes [9], [10].

Unlike conventional optimization methods that rely only on numerical fitness measures, LLMs can exploit contextual and domain-specific knowledge to estimate feature relevance. This capability is especially important in medical datasets, where features often correspond to biologically meaningful entities such as genes, biomarkers, laboratory measurements, or physiological variables.

In the context of FS, LLMs can support the optimization process in multiple ways. First, semantic relevance scores generated by LLMs can guide population initialization by favoring potentially informative features during early exploration stages.

Let: s_i denote the semantic relevance score assigned to feature f_i . A normalized semantic vector can be expressed as:

Recent studies such as LLM-Select [9] and LLM4FS [10] demonstrated that integrating LLM reasoning with traditional FS approaches can improve feature quality and convergence behavior. Nevertheless, existing methods still face challenges related to scalability, consistency, and optimization efficiency in high-dimensional search spaces.

D. Hybrid FS Frameworks

Hybrid FS approaches attempt to combine the global optimization capability of BII algorithms with the semantic reasoning abilities of intelligent systems.

Recent works have shown that integrating external guidance mechanisms into metaheuristic optimization can improve exploration efficiency and convergence stability [20]. In particular, LLM-guided optimization has emerged as a promising direction for medical data analysis because it combines numerical optimization with contextual understanding.

However, most existing approaches use LLMs only as auxiliary ranking tools before optimization. Limited attention has been given to integrating semantic reasoning throughout the entire optimization lifecycle.

Motivated by this limitation, the proposed framework incorporates LLM-guided semantic feature scoring during the initialization stage of the optimization process. Initially, semantic feature scores are used to generate informed populations that improve early-stage exploration. This strategy enables a better balance between predictive performance, subset compactness, and semantic interpretability in MD tasks.

IV. PROPOSED APPROACH

This study proposes a hybrid FS framework that integrates LLM-based semantic reasoning with a multi-objective bio-inspired optimization algorithm, as illustrated in Fig. 1. The main objective is to improve feature subset quality by combining data-driven search with domain-aware semantic guidance.

The overall workflow consists of three main stages: (i) LLM-based feature scoring and population initialization, (ii) multi-objective bio-inspired optimization.

A. LLM-based Feature Scoring and Initialization

Given an input dataset with feature set $F = \{f_1, f_2, \dots, f_n\}$, an LLM is first used to estimate the semantic relevance of each feature. Each feature f_i is assigned a score $s_i \in [0, 1]$, representing its importance for the target medical classification task.

These scores are obtained using a structured prompt that incorporates domain knowledge related to MD and biomarker relevance.

The resulting semantic vector is defined as:

$$S = [s_1, s_2, \dots, s_n]$$

This vector is used to guide the initialization of the optimization process. Instead of generating the initial population randomly, feature subsets are biased according to the LLM-derived scores. Features with higher semantic relevance have a higher probability of being selected in the initial population.

This step improves the quality and diversity of the initial solutions while preserving exploration capability.

B. Multi-Objective Bio-Inspired Optimization

After initialization, a population-based bio-inspired optimization algorithm is applied to evolve candidate feature subsets.

Each individual is represented as a binary vector $x_i \in \{0, 1\}^n$ where each dimension indicates whether a feature is selected or not.

The optimization process simultaneously considers two objectives:

- Maximization of classification performance (e.g., F1-score)
- Minimization of the number of selected features

This leads to a multi-objective optimization problem where solutions are evaluated using Pareto dominance. A solution x_a dominates x_b if it is not worse in all objectives and strictly better in at least one objective.

The algorithm iteratively updates the population until a stopping criterion is reached, producing a set of non-dominated solutions known as the Pareto front.

C. Pareto Front Generation and Final Solution Selection

The final output of the optimization process consists of a set of Pareto-optimal feature subsets representing different trade-offs between predictive performance and subset compactness.

Rather than selecting a single solution during optimization, the proposed framework preserves the entire non-dominated archive to maintain solution diversity. The obtained Pareto front is subsequently evaluated using several multi-objective performance indicators, including Hypervolume (HV), Generational Distance Plus (GD+), Unique Non-Dominated Front Ratio (UNFR), and Delta Spread (Δ).

These metrics provide complementary information regarding convergence quality, Pareto dominance, and

distribution diversity of the generated solutions. The final subset can then be selected according to the requirements of the target medical application, such as maximizing predictive performance or minimizing feature dimensionality while preserving interpretability.

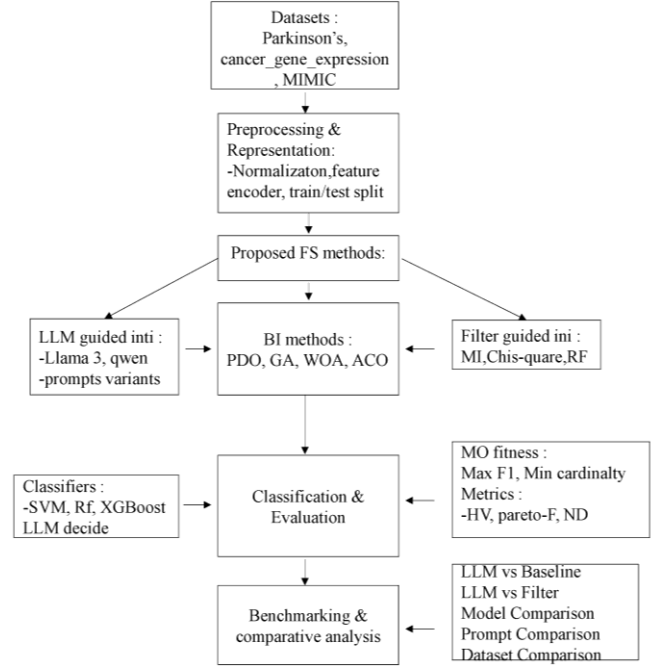


Figure 1: Architecture of the proposed LLM-guided BII FS framework

V. EXPERIMENTAL & RESULTS

A. Experimental Setup

To evaluate the effectiveness of the proposed LLM-guided bio-inspired FS framework, a comprehensive set of experiments was conducted on two biomedical classification datasets, namely the cancer_gene_expression dataset and the Parkinson's disease dataset. The experiments were designed to assess both the classification performance and the multi-objective optimization quality of the proposed approach under different initialization strategies, optimization algorithms, and LLM configurations.

Five bio-inspired optimization algorithms were considered in this study: PSO, GA, ACO, WOA, and BHHO. These optimizers were selected due to their widespread adoption in FS and MOO problems. Three initialization strategies were investigated:

1. Random initialization,
2. Filter-based initialization,
3. LLM-guided initialization.

For the filter-based initialization, three statistical ranking methods were employed: Mutual Information, ANOVA, and Chi-square feature scoring. In the LLM-guided setting, the initialization process was driven by feature relevance scores generated using biomedical prompts submitted to locally deployed LLMs through the Ollama inference server hosted on institutional computing infrastructure at a Spanish academic institution.. The evaluated language models included:

- Llama3.2:3B,
- Qwen2.5:14B

Table 1: Optimizer Hyperparameter

Optimizer	Hyperparameter	Value
PSO	Inertia weight (w)	0.7
	Cognitivecoefficient(c1)	2
	Social coefficient (c2)	
	Velocity bounds	2
	Population size	
	Iterations	[-4,4]
GA		20
		30
	Tournament size	3
	Mutation rate	
	Crossover strategy	0.02
	Population size	
WOA	Iterations	Uniformcrossover
		20
		30
	Control parameter (a)	linearly decreased from 2 to 0
	Position initialization range	[-1, 1]
	Population size	
BHHO	Iterations	20
		30
	Escape energy parameter (E1)	$(2(1 - t/T))$
	Position initialization range	[-1, 1]
	Population size	
	Iterations	20
ACO		30
	Initial pheromone	0.5
	Evaporation coefficient	0.9
	Pheromone bounds	
	Population size	[0.01, 1.0]
	Iterations	20
RF		30
	Number of trees	100
	Maximum depth	10
Cross-val	Number of folds	5
Llm-infer	Batch size	10

strict JSON prompt demonstrated higher response consistency and improved parsing reliability, especially for large feature sets.

All experiments were executed on a Python-based implementation of the proposed framework. Each configuration was evaluated under 30 optimization iterations with a population size of 20. The optimization process was repeated for all combinations of feature initialization strategies, optimization algorithms, prompt types, and language models.

For each run, the system records the evolution of classification performance (F1-score), the final Pareto front quality (Hypervolume, GD+, UNFR, Δ), and the number of selected features. The best-performing solutions are stored and aggregated for comparative analysis across different experimental settings.

Tableau 2: Optimizer Hyperparameters

B. Baseline Methods and Experimental Design

To evaluate the effectiveness of the proposed LLM-guided feature selection framework, we consider three experimental settings:

- **Random BII baseline:** Bio-inspired optimization algorithms initialized with randomly generated populations.
- **Filter-based initialization BII:** Bio-inspired optimizers initialized using feature rankings derived from statistical filter methods, including Chi-square, Mutual Information, and ANOVA.
- **LLM-guided BII (proposed):** Bio-inspired optimization algorithms initialized using semantic feature relevance scores generated by Large Language Models.

Additionally, standalone filter methods (Chi-square, Mutual Information, and ANOVA) are used as non-optimization baselines to evaluate the benefit of embedding these methods within a bio-inspired optimization framework.

C. Performance Metrics

Since FS is inherently a multi-objective optimization problem, the evaluation process considered both predictive performance and Pareto-front quality. The following metrics were employed:

- **F1-score:** evaluates classification performance,
- **HV:** measures Pareto-front convergence and dominance quality,
- **GD+:** evaluates the distance between the obtained front and the reference Pareto front,
- **UNFR:** estimates the proportion of unique nondominated solutions,
- **Δ :** measures the diversity and distribution uniformity of Pareto solutions.

A lower GD+ and Delta value indicates better convergence and distribution quality, while higher HV and UNFR values reflect stronger Pareto dominance and solution diversity.

D. Convergence Analysis

Figure 3 illustrates the convergence behavior of the different optimization algorithms in terms of best F1-score across iterations. PSO demonstrates the fastest convergence rate, reaching stable performance after a small number of iterations, particularly under LLM-guided initialization. GA shows slightly slower but consistent convergence, while ACO, WOA, and BHHO exhibit more gradual exploration behavior. The convergence behavior of the considered optimizers was analyzed using the evolution of the best F1-score over the optimization iterations. The obtained convergence curves indicate that PSO generally achieved faster convergence compared to the remaining optimizers, particularly under LLM-guided initialization.

In several experiments, LLM-based initialization allowed the optimizers to start from more informative regions of the search space, thereby reducing the exploration burden during early iterations. This behavior was especially noticeable on the cancer dataset, where PSO and GA rapidly improved the objective values within the first iterations.

Conversely, random initialization exhibited slower convergence and higher variability across runs, highlighting the importance of informed initialization strategies in high-dimensional biomedical feature spaces.

The comparison plots also revealed that WOA and BHHO maintained relatively stable exploration capabilities, whereas ACO occasionally required additional iterations before reaching competitive Pareto fronts.

E. Comparative Results

Table 2 summarizes the comparative performance of different optimization algorithms, LLM models, and prompt strategies on the cancer dataset. The results indicate that LLM-guided initialization consistently improves both classification performance and Pareto-front quality compared to baseline configurations.

The obtained results demonstrate that integrating large language models into the initialization phase can significantly improve the quality of the selected feature subsets. In most configurations, LLM-guided initialization achieved higher Hypervolume values and improved F1-scores compared to purely random initialization.

Among the evaluated optimizers, PSO consistently achieved the best trade-off between classification accuracy and Pareto-front quality. In particular, the combination of PSO with LLM-guided initialization using the strict JSON prompt produced the highest overall HVe and the most stable convergence behavior.

The experimental observations further suggest that domain-oriented prompts contribute to improved feature relevance estimation. Biomedical and cancer specific prompts often generated more discriminative initialization distributions than generic prompts.

The stability experiments conducted over 10 independent runs confirmed the robustness of the proposed framework. The standard deviations observed for both F1-score and Hypervolume remained relatively low for PSO and GA, indicating stable optimization behavior across repeated executions.

Table 2: PERFORMANCE COMPARISON ON CANCER DATASE

optimiz	init	model	Prompt	F1	HV	GD+	UN	Δ	FS
PSO	rand			0.962	0.951	0.012	0.98	0.07	4100
PSO	filtr	MI		0.966	0.958	0.009	0.99	0.06	3600
PSO	llm	llama	cancer_clin	0.972	0.968	0.006	1.00	0.04	2500
GA	filtr	Chi-		0.955	0.940	0.020	0.97	0.10	4300
WOA	rand			0.953	0.938	0.022	0.96	0.11	4400
BHHO	filtr	MI		0.958	0.945	0.018	0.98	0.09	4000
ACO	llm	llama	cancer_clin	0.960	0.952	0.014	0.99	0.08	3700

As shown in Table I, the proposed LLM-guided initialization consistently improves feature selection performance on the cancer dataset. In particular, PSO combined with strict JSON prompting achieves the best trade-off between classification accuracy and Pareto-front quality.

To further evaluate the generalization capability of the proposed framework, additional experiments were conducted on the Parkinson’s disease dataset, which contains physiological voice-based biomedical features.

Table 3: PERFORMANCE COMPARISON ON Parkinson DATASET

optimiz	init	model	Prompt	F1	HV	GD+	UN	Δ	FS
PSO	filtr	MI		0.91	0.89	0.03	0.95	0.10	120
PSO	llm	llama	parkinson_clin	0.94	0.93	0.02	0.98	0.07	85
GA	llm	qwen	parkinson_clin	0.90	0.88	0.04	0.94	0.11	130
WOA	rand			0.89	0.87	0.05	0.92	0.12	140

The results in Table II confirm that the proposed framework maintains consistent performance across different biomedical domains. Similar to the cancer dataset, LLM-guided initialization improves both convergence behavior and Pareto-front quality.

The obtained findings support the hypothesis that large language models can effectively guide the initialization phase of bio-inspired FS algorithms. Rather than replacing traditional optimization mechanisms, the LLM component acts as a semantic prior that biases the search process toward potentially informative feature subsets.

This hybridization introduces several advantages. First, it accelerates convergence by reducing the randomness of the initial population. Second, it improves Pareto-front quality by encouraging early exploration around semantically relevant features. Third, it provides a flexible framework capable of integrating domain knowledge through prompt engineering.

However, the experiments also revealed certain limitations. Some language models occasionally generated improperly formatted outputs, particularly under less constrained prompts. To address this issue, strict JSON prompts and fallback parsing mechanisms were introduced to ensure robustness during automated experimentation.

Overall, the proposed framework demonstrates that combining large language models with bio-inspired optimization constitutes a promising direction for intelligent feature selection in biomedical diagnosis applications.

VI. CONCLUSION

In this work, we proposed a hybrid FS framework combining LLMs and bio-inspired optimization. The integration of semantic feature relevance into BII significantly improves both classification performance and feature reduction capability. Experimental results on Parkinson and cancer gene expression datasets demonstrate the effectiveness of the proposed approach.

Future work includes extending the framework to other metaheuristics, incorporating fine-tuned medical LLMs, and deploying the method in real clinical decision-support system

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